Toward Multimodal Agents that Perceive, Reason, Act, and Predict

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April 30, 2026

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Past Work: Learning to Perceive, Reason, and Act through Tools with Multimodal RL

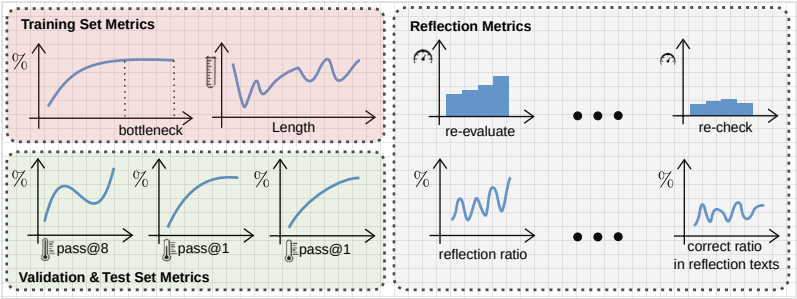
Starting with Visual Reasoning RL

I use visual reasoning RL as a first step toward improving the reasoning capability of multimodal agents under current observations.

First question: how can we make claims about multimodal reasoning RL trustworthy?

How Can We Make Claims About Multimodal Reasoning RL Trustworthy?

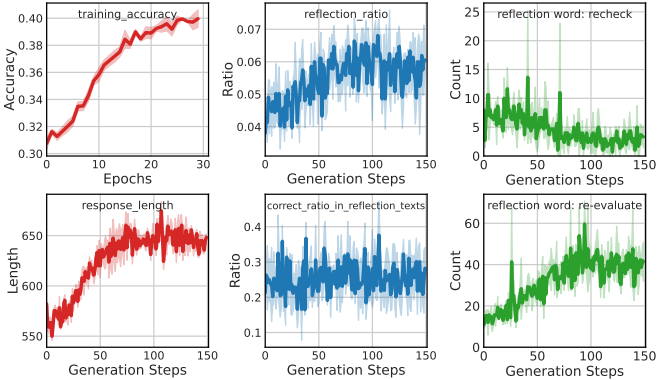
Early multimodal RL was hard to interpret from final tables and heterogeneous codebases. I built a transparent from-scratch VLM-RL framework and evaluated full learning trajectories on visual reasoning tasks.



[2025.01-2025.03] *Rethinking RL Scaling for Vision Language Models: A Transparent, From-Scratch Framework and Comprehensive Evaluation Scheme*, Ma et al.

What Does RL Change During Multimodal Reasoning Training?

Trajectory-level evaluation reveals that RL changes how VLMs learn and respond, not only where final accuracy lands.

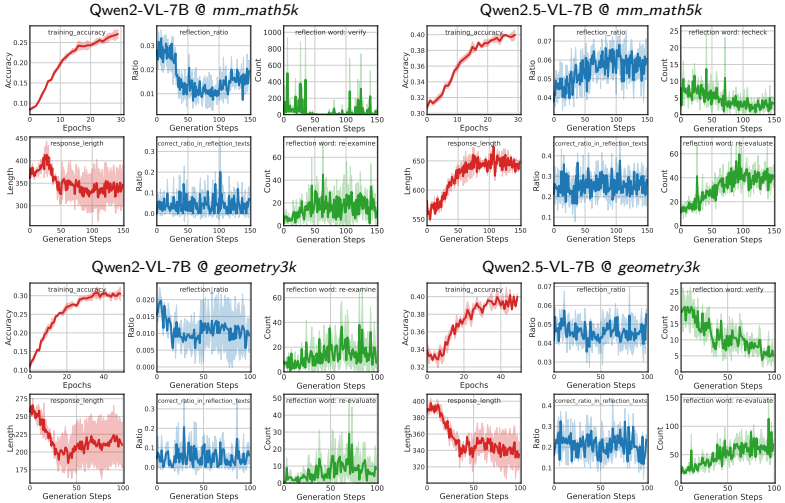


Accuracy improves | Length shifts | Reflection changes | Seeds matter

[2025.01-2025.03] *Rethinking RL Scaling for Vision Language Models: A Transparent, From-Scratch Framework and Comprehensive Evaluation Scheme*, Ma et al.

Do Multimodal RL Dynamics Look the Same Across Settings?

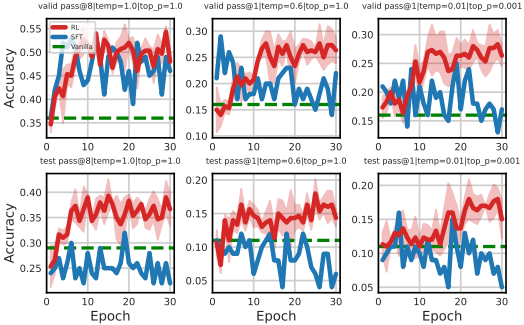
The same multimodal RL evaluation scheme reveals different dynamics across backbones and datasets.



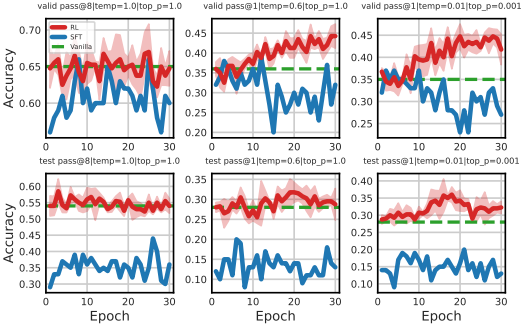
Does Multimodal RL Change Generalization Beyond SFT?

Compared with SFT on high-quality CoT solutions from textbooks, RL shows stronger validation/test generalization.

Qwen2-VL-7B @ *mm_math5k*



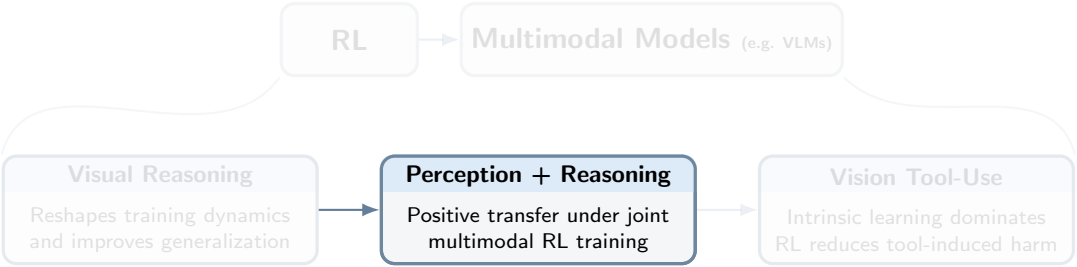
Qwen2.5-VL-7B @ *mm_math5k*



RL improves generalization where SFT can overfit to supervised reasoning traces.

From Visual Reasoning to Perception + Reasoning

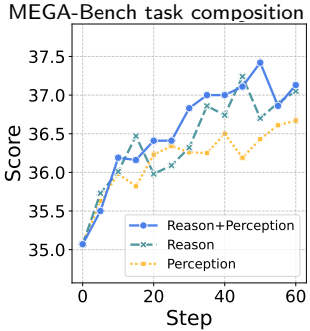
Reasoning decides how to think; perception determines whether the model is thinking about the right visual evidence.



Next question: can multimodal RL improve both within one training pipeline?

Can Multimodal RL Jointly Shape Perception and Reasoning?

Under matched budgets, unified RL matches or outperforms specialist mixtures across both reasoning and perception.



	Reasoning Benchmarks				
Mixture	MMMU	MathVista	MathVision	MME-R	CharXiv (RQ)
Unified	56.56	71.40	30.51	**28.20**	**44.50**
Reason-only	54.89	**71.70**	**31.24**	28.03	44.00
Perception-only	**56.67**	68.20	29.59	26.26	41.00

	Perception Benchmarks				
Mixture	HrBench4K	VStar	COCO (S|M)	OCRBenchV2	ScreenSpot-Pro
Unified	73.75	**82.20**	80.69 | **38.62**	**55.87**	23.91
Reason-only	73.25	78.53	78.36 | 33.87	55.77	**23.97**
Perception-only	**76.00**	80.10	**80.78** | 37.37	55.33	23.78

Reasoning: math / science / chart / puzzle

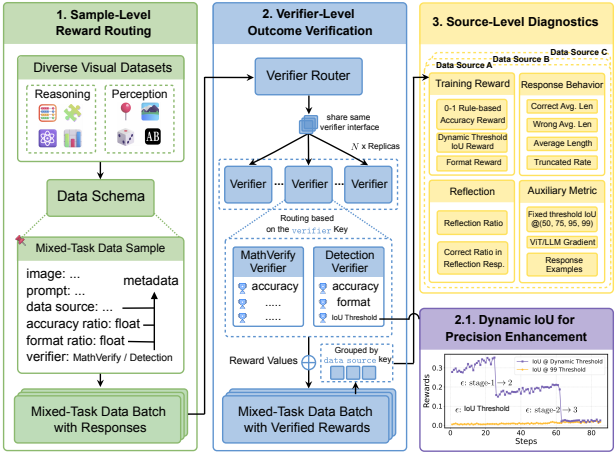
Perception: detection / grounding / OCR / counting

47.7K verifiable samples

Qwen2.5-VL 7B / 32B-0321/0326

[2025.04-2025.06] *One RL to See Them All: Visual Triple Unified Reinforcement Learning*, Ma et al.

How Can Unified Multimodal RL Handle Heterogeneous Tasks?



We make heterogeneous multimodal tasks trainable in one RL loop.

Reward routing
handles sample-level heterogeneity

↓

Outcome verification
handles reward-side heterogeneity
Dynamic IoU for localization

↓

Source-level diagnostics
reveals source-specific behaviors and failures

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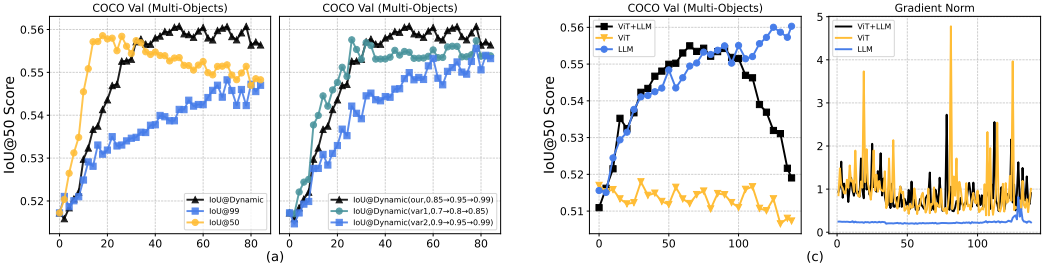
What Made Unified Multimodal RL Stable in Practice?

In unified multimodal RL, stability bottlenecks arise from both sparse localization rewards and fragile vision-model updates.

Fixed IoU: loose is ambiguous, strict is sparse

Updating ViT causes gradient spikes



Dynamic IoU: loose-to-strict localization curriculum

Freezing ViT prevents perception degradation

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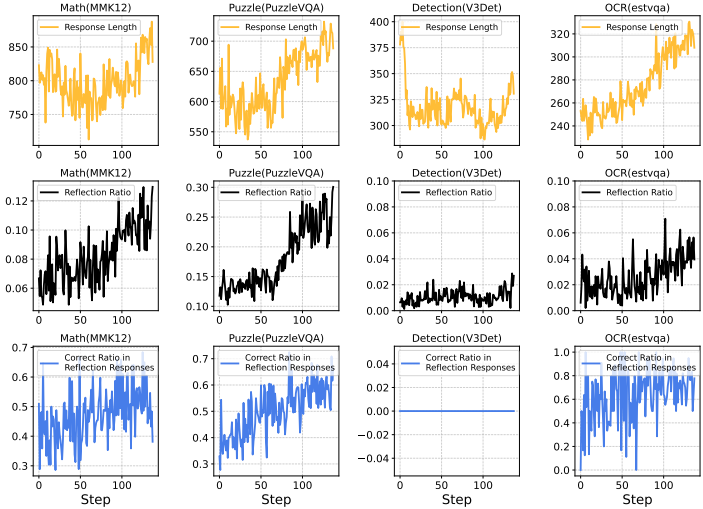
What Made Unified Multimodal RL Diagnosable?



Source diagnostics reveal task-specific behavior patterns hidden by aggregate metrics.

Reasoning sources show length and reflection growth

Detection and OCR keep low-reflection patterns

Unified RL training preserves task-appropriate response styles

Can Unified Multimodal RL Scale with New Task Domains?

Under a fixed-compute budget, unified training shows positive transfer when adding a new GUI grounding domain.

Mixture	ScreenSpot-Pro	OCRBenchV2
Unified	23.91	55.87
Perception-only	23.78	55.33
Perception-only + GUI-3K	29.85	55.96
Unified + GUI-3K	**31.68**	**56.09**

New-domain data improves ScreenSpot-Pro while OCRBenchV2 is preserved or slightly improved.

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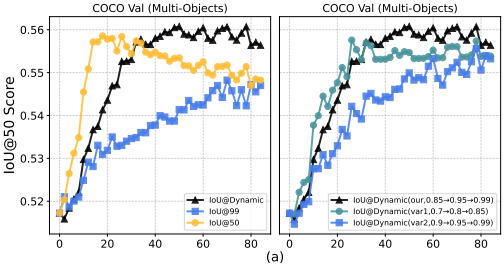

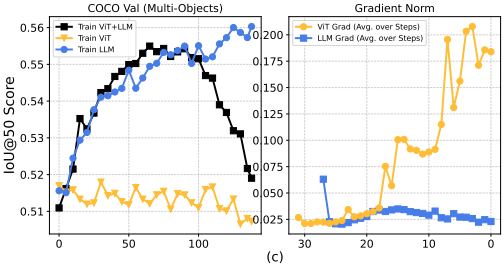

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